

Development Roadmap:

Week	Development Phase	Major Tasks	Responsible Members	Expected Completion
Week 06	Backend Foundation	Set up Node.js and Express.js, connect MongoDB Atlas, create database models, and implement authentication and profile APIs	Suhani, Musa	Working backend, database connection, registration, login, and profile APIs
Week 07	Frontend Development	Set up React.js and Tailwind CSS, create routing, and implement Login, Registration, Dashboard, and Profile pages	Sabequn, Musa	Core frontend pages connected with authentication and profile APIs
Week 08	AI Integration	Connect Gemini API, prepare structured prompts, and implement workout and nutrition recommendation endpoints	Arpita, Suhani	AI-generated workout and nutrition recommendations working through the backend
Week 09	Feature Completion	Implement progress tracking, exercise library, recommendation history, categories, and Admin Dashboard functions	All members	All major user and admin features completed
Week 10	System Integration	Connect remaining frontend pages with backend APIs and verify complete data flow between React, Express, MongoDB, and Gemini	All members	Fully integrated application
Week 11	Testing and Debugging	Perform functional, API, authentication, responsive design, error-handling, and AI-output testing	All members	Major bugs identified and corrected
Week 12	Deployment	Deploy frontend on Vercel and backend on Render, configure MongoDB Atlas and environment variables, and test the deployed system	Musa, Suhani	Application successfully deployed online
Week 13	Documentation and Improvement	Update README, setup instructions, API documentation, screenshots, test results,	Musa, Arpita, Sabequn	Final documentation and polished application

		and make final UI improvements		
Week 14	Final Review and Presentation	Conduct final system testing, prepare demonstration and presentation materials, and complete final submission	All members	Project ready for evaluation and presentation

Team Task Distribution:

Team Member	Official Role	Assigned Responsibilities	Expected Deliverables
Musa Islam Shadin	Team Lead	Manage the development schedule and team coordination; maintain GitHub repository and branches; supervise frontend-backend integration; review system architecture and requirements; coordinate testing and debugging; manage deployment; prepare README, reports, diagrams, and final presentation	Organized repository, integrated system, deployment, updated documentation, testing report, and final presentation
Arpita Sarker	AI Integration Lead	Prepare structured prompts for workout, nutrition, and progress recommendations; define AI input and output formats; configure and verify Gemini API responses with the backend developer; test recommendations using different user profiles, goals, and budgets; check AI responses for relevance and safety; document the AI workflow and prepare the AI module explanation for presentation	Prompt templates, tested AI recommendations, AI response structure, AI workflow documentation, and AI module presentation
Easrin Mahjabin Suhani	Backend Developer	Set up Node.js and Express.js; design MongoDB models; implement authentication, profile, progress, exercise, category, and recommendation APIs; connect MongoDB Atlas; implement JWT and bcrypt security; connect Gemini API endpoints with Arpita; perform API testing; deploy the backend on Render	Working backend server, database models, REST APIs, authentication, Gemini endpoints, API test results, and deployed backend

Sabequn Nahar	Frontend Developer	Set up React.js and Tailwind CSS; convert Figma designs into responsive pages; implement routing, forms, navigation, dashboards, and loading/error states; connect frontend pages with backend APIs; display AI recommendations and progress information; perform responsive testing; deploy the frontend on Vercel	Responsive frontend, connected forms and pages, AI result display, frontend test results, and deployed frontend
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Technology Setup Plan:

Component	Selected Technology	Purpose
Frontend	React.js	Develop the interactive web interface
Styling	Tailwind CSS	Create a responsive and consistent design
Backend	Node.js and Express.js	Develop REST APIs and handle system logic
Database	MongoDB Atlas	Store users, profiles, progress records, exercises, categories, and recommendations
AI Service	Gemini API	Generate workout guidance, nutrition suggestions, and progress feedback
Authentication	JWT and bcrypt	Manage secure authentication and password hashing
Frontend Deployment	Vercel	Host the React frontend
Backend Deployment	Render	Host the Node.js and Express.js backend
Version Control	Git and GitHub	Manage source code and team collaboration
API Testing	Postman	Test backend endpoints and responses
Code Editor	Visual Studio Code	Develop and manage the project code
Package Management	npm	Install and manage project dependencies

Design and Diagram Tools:

Tool	Usage
Figma	UI/UX design and screen prototyping
Draw.io	User flow and system diagrams
Microsoft Word	Project documentation
GitHub	Documentation and source-code management